

China's, but lags by about five years," Thakur says.

Where investors see the next big wave differs by sector but aligns on a few themes: AI-enabled genomic interpretation, population-scale sequencing and rare-disease diagnostics. Companies like Vgenomics, building AI-driven interpretation engines, say automation will fill the gap created by India's shortage of genetic counsellors and bioinformaticians.

Public projects such as Genome India and the Bharat Cancer Genome Atlas are expected to enrich India-specific datasets, enabling more accurate variant interpretation and opening new commercial pathways in pharmacogenomics

hope that India will become a major sequencing market in Asia.

INSIDE THE LAB

Without the massive infrastructure build-out underway across labs, hospitals and sequencing companies, India's genomics push would not be viable. That build-out is accelerating on multiple fronts: data generation, sequencing throughput, automation, bioinformatics and clinical integration.

"Diagnostics in India has mostly been reactive," says Surajit Chakrabartty, the Chief Financial Officer of MedGenome. "Genomic testing sits at the highest end of the diagnostic spectrum," he says, noting that awareness, affordability,

clinician-prescribed testing, supported by partnerships with 8,000 hospitals and 24,000 clinicians. The company recently raised \$47.5 million. AI is vital to its workflow.

"Our bioinformatics tools help identify population-specific variants and improve clinical relevance for the Indian population," Chakrabartty says.

Metropolis Healthcare is placing high-end genomics at the core of its growth strategy. "Genomic testing in India has shifted from a specialised offering to a key driver of personalised medicine," says Surendran Chemmenkotil, Managing Director. Its oncology, reproductive genomics and rare-disease segments are seeing steep demand

66 |

CLINICAL GENOMICS HAS MOVED FROM THE PERIPHERY TO THE CORE... DEMAND IS RISING IN TANDEM ACROSS ALL MAJOR SEGMENTS LIKE ONCOLOGY & RARE DISEASES

and precision therapeutics.

Despite momentum building, risks remain. Thakur of Quadria Capital outlines the five constraints every investor maps: testing costs still ranging between ₹20,000 and ₹50,000; insurance reimbursement is limited; national genomic databases are still inadequate; the regulatory frameworks are evolving; and there is a shortage of specialised talent.

"We support companies with AI-driven interpretation and hospital networks that can bring prices under ₹10,000 per test," he says. Yet the sector continues to attract capital.

Quadria Capital and HealthQuad have doubled down on genomics, illustrating a long-term

and access have historically been the biggest barriers.

Yet, demand is rising. Oncology and rare-disease testing lead the surge, while preventive genomics boosted by younger consumers is emerging rapidly. "...but penetration is low. Growth remains strong with significant untapped potential," he says.

MedGenome runs a 30,000-sq-ft multiomics laboratory in Bengaluru, one of the largest of its kind in South Asia. It has processed over 500,000 genomes and exomes and serves more than a million patients. Its high-throughput sequencers, bioinformatics engines, and AI-driven interpretation workflows aid many hospital-driven tests.

Most of its revenue comes from

curves. The company has committed ₹20 crore over three years to expand sequencing capacity.

Metropolis now sees genomics growing three times faster than its broader diagnostics portfolio. Although it currently contributes a single-digit percentage of revenue, it is already at break-even.

Strand Life Sciences is scaling capacity through its new Genomic Diagnostics & Research Center in Bengaluru. The facility is designed to triple growth.

Illumina, the global sequencing company, is expanding access to advanced sequencing across India, including Tier-II and Tier-III cities. "In India, non-communicable diseases account for two-thirds of deaths," says Suchita Dayanand,